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The image shows a JupyterLab notebook titled "Day3_Python_Test_Solution" with a last checkpoint 32 minutes ago. The notebook is running on Python 3 (ipykernel). The code is divided into three sections:

Q.1) Write a code to Read a file and append lines to a list.

```
[13]: lines_list = []
with open("test.txt", "r") as file:
    for line in file:
        lines_list.append(line.strip())

print(lines_list)

['my name is Saurabh', 'i have many friends']
```

Q.2) Write a code to catch an Exception in python?

```
[14]: try:
    x = int(input("Enter number: "))
    y = int(input("Enter number: "))
    print(x / y)
except Exception as e:
    print("Error:", e)

Enter number: Saurabh
Error: invalid literal for int() with base 10: 'Saurabh'
```

Q.3) Write a Python function that accepts a list containing strings and integers. Merge all string elements using # and add all integer elements. e.g. input list is ['100', 'welcome', 'hi', '200', '300', 'bye', 'welldone', '500'] Output should be: welcome#hi#bye#welldone# 1100

```
[15]: lst = ['100', 'welcome', 'hi', '200', '300', 'bye', 'welldone', '500']
s = ""
total = 0
for i in lst:
    if i.isdigit():
        total += int(i)
    else:
        s += i + "#"
print(s)
print(total)

welcome#hi#bye#welldone#
1100
```

Q.4) Write a script to sort a dictionary based on its values and find the sum of middle two values

```
input_dict = {"x": 5, "y": 15, "z": 25}
Output:
Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}
Sum of middle two values: 15 + 5 = 20
or
input_dict = {"x": 5, "y": 15, "z": 25}
sorted_dict = sorted(input_dict.items(), key=lambda item: item[1])
print(sorted_dict)
[('x', 5), ('y', 15), ('z', 25)]
sum_of_middle_two_values = sorted_dict[1][1] + sorted_dict[0][1]
print(sum_of_middle_two_values)
20
```

Day3_Python_Test_Solution

localhost8889/notebooks/Day3_Python_Test_Solution.ipynb

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JupyterLab Python 3 (ipykernel)

Q.4) Write a script to sort a dictionary based on its values and find the sum of middle two values

values

input_dict = {"x": 5, "y": 15, "z": 25}

Output:

Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}

Sum of middle two values: 15 + 5 = 20

or

input_dict = {"x": 5, "y": 15, "z": 25, "p": 12}

Output:

Sorted Dictionary: {'x': 5, 'p': 12, 'y': 15, 'z': 25}

Sum of middle two values: 12 + 15 = 27

```
[22]: def middle_sum(d):
    if not d:
        print("Dictionary is empty. Cannot calculate sum.")
        return
    sorted_items = dict(sorted(d.items(), key=lambda x: x[1]))
    values = list(sorted_items.values())
    n = len(values)

    if n == 1:
        s = values[0] * 2
    elif n % 2 == 0:
        s = values[n//2 - 1] + values[n//2]
    else:
        s = values[n//2] * 2
```

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JupyterLab Python 3 (ipykernel)

```
sorted_items = dict(sorted(d.items(), key=lambda x: x[1]))
values = list(sorted_items.values())
n = len(values)

if n == 1:
    s = values[0] * 2
elif n % 2 == 0:
    s = values[n//2 - 1] + values[n//2]
else:
    s = values[n//2] * 2

print("Sorted Dictionary:", sorted_items)
print("Sum =", s)
middle_sum({"x":5,"y":15,"z":25})
middle_sum({"x":5,"p":12,"y":15,"z":25})
middle_sum({"a":10})
middle_sum({})
```

Sorted Dictionary: {'x': 5, 'y': 15, 'z': 25}

Sum = 30

Sorted Dictionary: {'x': 5, 'p': 12, 'y': 15, 'z': 25}

Sum = 27

Sorted Dictionary: {'a': 10}

Sum = 20

Dictionary is empty. Cannot calculate sum.

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